Endocrine Disorders, Fertility, and Sexual Health

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56.1 Endocrine Function

Endocrine dysfunction is frequent in long-term survivors after HCT, particularly in children with a global rate of 58.7% with at least one endocrine abnormality (Güemes et al. 2022). Underlying disease, pretransplant therapy, age at HCT, type of conditioning, total body irradiation, development of chronic GVHD, and its treatment can all contribute to the risk of endocrine dysfunction. Consideration of pretransplant risk factors and their interaction with HCT-related exposure should be considered to assess the risk of endocrine dysfunction. Since many long-term complications may not manifest for years or even decades after HCT, survivors need ongoing, lifelong monitoring (Bhatia et al. 2017). Systematic follow-up is important to identify and treat endocrine defects before clinical impact, and this is particularly important in children where growth and puberty are at risk.

56.1.1 Thyroid Dysfunction

56.1.1.1 Background

Thyroid dysfunction is a well-recognized late complication after HCT. The most common abnormality of thyroid function after HCT is primary compensated hypothyroidism. Approximately one in four patients develops hypothyroidism after allogeneic HCT, with a greater incidence in females (Felicetti et al. 2023). This may not require treatment and commonly resolves. Overt hypothyroidism may be primary or less commonly central. Other thyroid disorders include autoimmune thyroid disease (thyroiditis, Graves' disease) and thyroid cancers (carcinomas or benign adenomas). In a single-center study of 791 patients followed up for 38 years, new cases of thyroid dysfunction continued for 28 years after HCT highlighting the need for indefinite follow-up (Sanders et al. 2009).

Risk factors for hypothyroidism post-HCT include the use of TBI with single dose being associated with a fivefold to sixfold higher risk than fractionated TBI. BU-based regimens are more likely to cause thyroid problems than those containing CY only, and patients with malignant disease (e.g., Hodgkin lymphoma) are more likely to get thyroid dysfunction than patients

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with non-malignant diseases (e.g., aplastic anemia). The age of the patient is also important with younger patients at higher risk (Cohen et al. 2007). In adults, recent data show that advanced age at transplantation is associated with higher risk for hypothyroidism (Ataca Atilla et al. 2020). Pre-transplant TSH levels may predict the onset of post-HCT hypothyroidism (Felicetti et al. 2023). Thyroid dysfunction is more likely to occur in patients receiving prolonged immunosuppression for chronic GVHD (Savani et al. 2009). Furthermore, the risk of developing hypothyroidism is higher in patients treated with multiple allo-HCTs who have been transplanted for acute leukemia (Medinger et al. 2017). In relation to thyroid cancer, a retrospective study which included data on >68,000 patients showed that the relative risk (RR) of thyroid cancer was approximately threefold higher following HCT than in the general population. The RR was >20 if transplanted before the age of 10 years and close to 5 if transplanted between age 11 and 20 years. Female gender and GVHD were additional risk factors (Cohen et al. 2007).

56.1.1.2 Prevention/Management

Continuous and long-term monitoring of thyroid function after HCT is essential for early diagnosis and to provide timely and appropriate treatment. Patients should have annual laboratory assessment with thyroid-stimulating hormone (TSH) and free thyroxine (FT4) with more frequent testing if there is clinical suspicion of thyroid dysfunction. Annual clinical examination should include palpation of the thyroid gland, and there should be a low threshold for arranging a thyroid ultrasound (Bhatia et al. 2017).

56.1.2 Hypoadrenalism

56.1.2.1 Background

The main risk factor is the use of glucocorticoids which lead to central corticotrophin deficiency. Patients are at risk if they have received supraphysiological doses for 1 month or more, and topical, inhaled, intranasal, oral, and injectable forms all pose a risk (Gurnell et al. 2021). TBI can also cause corticotrophin deficiency as can drugs. Diagnosis can be difficult as many patients will have few if any symptoms or they may present with nonspecific symptoms such as fatigue, weakness, nausea, weight loss, and hypotension. Some symptoms may mimic GVHD. Identifying patients who are at higher risk is helpful in guiding the diagnosis. Diagnosis requires paired morning cortisol and ACTH levels. If the results are inconclusive, then additional investigations should be arranged with an endocrinologist.

56.1.2.2 Prevention/Management

When hypoadrenalism is confirmed, hydrocortisone should be given with additional doses to cover stresses such as illness, infection, or surgery. Subsequently, regular evaluation is required as it may be possible to reduce/stop medication (Cornillon et al. 2013).

56.1.3 Growth

56.1.3.1 Background

Short stature is multifactorial after transplant. It is a recognized side effect of radiation to the hypothalamic-pituitary area given in childhood due to a reduction in growth hormone (GH) secretion. Radiation can also induce bone lesions. Pretransplant cranial radiation (e.g., patients with ALL) is also relevant, and single-dose TBI rather than fractionated radiation increases the risk further.

Additional contributory factors to short stature in these patients include underlying disease (e.g., Fanconi anemia), other hormone deficiencies (including thyroid and gonadal hormones), nutritional deficits, illness, steroids, and GVHD. Male sex and young age at time of transplant are additional risk factors.

56.1.3.2 Prevention/Management

Children's growth velocity should be closely monitored with height and weight documented at each clinic visit. A possible increased risk of secondary malignancies has been described in patients receiving growth hormone (GH) replacement therapy after previous neoplasia; this has

raised concerns regarding the use of GH in the absence of sufficient long-term follow-up data. As a consequence of this, there are currently no clear guidelines for the use of GH in these patients. A pediatric endocrinologist should be involved if growth rate is abnormal based on bone age and pubertal stage (Chow et al. 2016; Lawitschka et al. 2019), and the use of GH therapy should be considered for children whose height standard deviation score is less than 2.

56.2 Gonadal Dysfunction and Infertility

56.2.1 Background

Normal reproduction in both sexes requires germ cells and an intact hypothalamic-pituitary endocrine axis. In female patients, the uterus must be both receptive to implantation and capable of undergoing growth during pregnancy. Chemotherapy and radiation can lead to damage in all of these areas and compromise the likelihood of successful parenthood after HCT. Before starting any chemoradiotherapy regimen, the potential effects on the future fertility of the patient should be considered and discussed with the patient together with a discussion of fertility-preserving strategies and depending on the country may require a dedicated interview with a reproductive medicine specialized physician.

56.2.2 Gonadal Dysfunction in Women Following Chemoradiotherapy

Women are born with a finite number of eggs which can be fertilized for pregnancy or depleted over time as a result of physiological apoptosis or else menstruation. Chemoradiotherapy depletes further the number of follicles by (1) activating apoptotic pathways, (2) causing fibrosis of stromal blood vessels, and (3) activating resting (antral) follicles, leading to a "burnout" effect (Meirow and Nugent 2001; Kalich-Philosoph et al. 2013). The degree of ovarian damage is related to the dose and type of chemotherapeutic agent used and baseline ovarian reserve which in turn is dependent on age and previous treatment. Manifestations of premature ovarian failure range from premature menopause to varying degrees of infertility. Alkylating agents have the highest age-adjusted odds ratio of ovarian failure (Meirow 2000). Even though some pregnancies and live births have been reported after either chemo-based or TBI-based myelo-ablative conditioning regimens, the risk of infertility after such MAC is very high. A combination of BU and CY is particularly gonadotoxic to females, but younger patients who receive CY only may have some gonadal function preserved, and pregnancies following CY are well described (Salooja et al. 2001).

Being transplanted in the peri-pubertal period with full-dose BU does represent a very high risk of infertility. Considering serum gonadotrophin level as a marker of ovarian failure, TREO seems to be less toxic than BU. However, we do not yet have evidence that pregnancy rates are higher in young female patients receiving TREO compared to those receiving BU (Faraci et al. 2019). TBI is also potentially sterilizing. The estimated median lethal dose of radiation for the human oocyte is less than 2 Gy (Wallace et al. 2003a). The effective sterilizing dose (ESD) decreases with increasing age, and while estimated as 18.4 Gy at 10 years of age, the ESD is approximately 14.3 Gy at 30 years of age and only 6 Gy in women over age 40 (Wallace et al. 2003b).

56.2.3 Gonadal Dysfunction in Men Following Chemoradiotherapy

In male patients, spermatogenesis is frequently impaired following chemoradiotherapy, but testosterone levels generally remain normal because of the relative resistance of testosterone producing Leydig cells to chemoradiotherapy. As a result, secondary sexual characteristics remain normal for male patients, and typically testosterone levels and luteinizing hormone (LH) levels are in the normal range. However, testosterone

levels have to be monitored on an annual or biannual basis since some male patients do experience hormonal deficiency, and this is associated with a reduction in quality of life. Spermatogonia are very sensitive to irradiation, and it takes approximately 2 years for sperm counts to recover to pre-irradiation levels after a single dose of 1 Gy (Meistrich and van Beek 1990). With higher doses, azoospermia persists longer or may be permanent. Following HCT conditioned with myeloablative doses of TBI, the majority of men will be azoospermic. Chemotherapy-only regimens are also associated with azoospermia but to a lesser degree (Rovo et al. 2013). Following BU, for example, approximately 50% will be azoospermic, while after CY alone recovery of spermatogenesis is more frequent (Mathiesen et al. 2020, 2021, 2022).

56.2.4 Uterine Dysfunction in Women After Radiation

Uterine development commences at puberty and is associated with an increase in both size and vascularity (Laursen et al. 1996). Exposure to radiation leads to reduced vascularity, fibrosis, and hormone-dependent endometrial insufficiency, which subsequently lead to adverse reproductive outcomes. Increased rates of infertility, miscarriage, preterm labor, intrauterine growth retardation, and low newborn birth weight have been described (Reulen et al. 2009), particularly if conception occurred within a year of radiotherapy (Fenig et al. 2001). Recently, uterine damage after BU has been reported (Courbiere et al. 2023).

56.2.5 Prevention/Management of Gonadal Failure

56.2.5.1 Fertility Preservation in Males

Sperm cryopreservation is an established fertility preservation option for postpubertal boys and men. Sperm can be used either for artificial insemination or, if the quantity and/or quality of sperm are insufficient, for intracytoplasmic sperm injections for in vitro fertilization. There is a chance of sperm recovery with time particularly if the patient was under the age of 25 years at transplant, did not have TBI, and has no evidence of chronic GVHD (Rovo et al. 2013). These patients require reassessment at intervals to ascertain their fertility potential. Prior to transplant some patients who are unable to ejaculate will require interventions such as penile vibratory stimulation or else electroejaculation. If these fail, testicular sperm extraction (TESE) can be used to obtain samples for cryopreservation. (Halpern et al. 2020).

Testicular tissue cryopreservation in prepubertal male patient represents a feasible but experimental technique since hitherto there has been no demonstration of obtaining vital sperm from the testis sample in vitro or in vivo in humans. Some animal models including nonhuman primates are encouraging (Kanbar et al. 2022).

56.2.5.2 Fertility Preservation Techniques in Females

Gonadotropin-Releasing Hormone Agonists (GnRHa)

Despite success in animal models, the value of GnRHa to preserve ovarian function during chemotherapy in human subjects is uncertain. A Cochrane database review concluded that the use of GnRH agonists should be considered for ovarian protection in women of reproductive age who are receiving chemotherapy (Chen et al. 2011).

Embryo and Oocyte Cryopreservation

Embryo and oocyte cryopreservation are preferred methods of fertility preservation in women who require sterilizing treatment. The use of donor embryos/oocytes can also be discussed with the patient because they offer the possibility of pregnancy and parenthood albeit with a nongenetic child. Mature oocyte collection requires ovarian stimulation. These oocytes can then either be frozen or else fertilized

in vitro before freezing. These options are not open to all patients however. Ovarian stimulation takes a minimum of 2 weeks, and this delay is prohibitive for many patients with hematological malignancies. The requirement for a partner or donor sperm for embryo cryopreservation is another potential drawback; for some patients, sperm is not available, and for others, the involvement of a partner/sperm donor limits future reproductive autonomy as consent from the sperm provider must be given not only at the time of cryopreservation but also at the time of reimplantation.

Ovarian Tissue Cryopreservation (OTC)

OTC is no longer considered experimental by the American Society for Reproductive Medicine Practice Committee (2019), but it remains unavailable to many patients. It is the only option open, however, to prepubertal patients or to women who cannot tolerate a significant delay in treatment due to disease severity or progression. Cortical fragments containing primordial follicles with immature oocytes can be obtained by laparoscopy and cryopreserved. Ideally, ovarian tissue should be obtained before the patient has been exposed to chemotherapy, but this is not always possible and is not an absolute requirement.

A major concern reimplanting cryopreserved ovarian tissue is the possibility of reseeding the tumor. The risk depends on the individual disease. Assessment by PCR of ovarian tissue taken from patients with leukemia (CML, AML, ALL), tested positive for disease in a number of cases and assessment of tissue from mice with severe combined immunodeficiency confirmed the leukemic potential of the tissue (Rosendahl et al. 2013). As a result, reintroduction of ovarian tissue from patients with leukemia would not currently be recommended. In the future, maturation in vitro of follicles from cryopreserved tissue may enable production of a viable disease-free alternative. In patients with lymphoma, histologically negative samples of ovarian issue have been transplanted without initiating relapse, but in some cases the follow-up time was short.

56.2.5.3 Children and Adolescents

Fertility preservation in children has been the subject of recent guidelines from the pediatric diseases working party of the EBMT (Dalle et al. 2017; Balduzzi et al. 2017). Extreme sensitivity is required, and parents have to be given complete information on the process, associated risks, and success rates. For prepubertal girls, OTC is currently the only potential fertility-sparing option. In peri-pubertal boys, it is sometimes possible to extract sperm using surgery/microdissection or electroejaculation under general anesthetic. In prepubertal males, the only option is testicular tissue cryopreservation; although work in animal models is encouraging, there have been no reports to date of reimplanted testicular tissue leading to human live births.

56.2.6 Management of Pregnancy After HCT

Most patients or their partners who conceive after HCT have uncomplicated pregnancies. Chemoradiotherapy can potentially affect a variety of maternal organs relevant to a successful pregnancy outcome, for example, renal, cardiac, and pulmonary toxicity. Patients at risk should have an expert medical review early in pregnancy and may require regular specialist monitoring throughout and review by an anesthetist prior to delivery. Patients who have had TBI or pelvic irradiation have an increased risk of premature and small birth weight babies and may be at increased risk of miscarriage. In the absence of TBI, miscarriage rates are typically comparable to the background population, and no significant increase in congenital malformations or genetic abnormalities has so far been described when conception has taken place long after completion of therapy (Meirow and Schiff 2005; Green et al. 2009). Animal experiments suggest that most cytotoxic drugs are mutagenic and teratogenic to oocytes exposed during the maturation phase. In humans, this phase lasts approximately 6 months (Meirow and Schiff 2005), so there is a theoretical advantage to delaying conception for 6 months after completing gonadotoxic treatments.

56.3 Sexual Function

56.3.1 Background

Sexual dysfunction is one of the most frequently reported complications after HCT. It has a significant impact on quality of life, and it is important to patients, ranking among their top unmet needs (Bevans et al. 2017; Schover et al. 2014). In a study including 1742 survivors of transplant with a mean follow-up of 11.9 years, 40% of women and 27% of men had not been sexually active in the previous year, while 64% of women and 32% of men who were sexually active reported low sexual function (Syrjala et al. 2021).

Complications affect recipients of both allogeneic and autologous transplants and include decreased libido, dyspareunia ± vaginal dryness (females), and erectile and ejaculatory dysfunction (males) (Li et al. 2015). Allogeneic recipients have additional problems linked to acute or chronic GVHD (Wong et al. 2013). Sexuality is also affected, and this is multifactorial due to decreased self-confidence, stress, anxiety, and fear of recurrence, together with a change in body image (Yi and Syrjala 2009).

A number of medical conditions can impact on sexual health after HCT, for example, genital changes, hormone changes, vascular disease, or other chronic illness. The sexual well-being of the survivor is also determined by their relationship with their partner who may experience a decrease in sexual desire and anxiety about initiating sexual activity with their survivor partner (Langer et al. 2007).

56.3.2 Prevention/Management

It is important to identify relevant issues before problems with sexuality and intimacy become entrenched. Furthermore, discussion with a health-care professional (HCP) may be associated with a reduction in the development of sexual problems (El-Jawahri et al. 2018; Humphreys et al. 2007). Current guidelines endorse the importance of discussion including both transplant follow-up guidelines (Majhail et al. 2012) and cancer survivorship guidelines from ASCO and NCCN (Carter et al. 2018; Denlinger et al. 2017). Transplant guidelines recommend regular discussion of sexual function at 6 months, 1 year, and annually thereafter while guidelines from NCCN and ASCO recommend that discussion is initiated during treatment planning. The patient experience, however, is that discussion happens infrequently (Gjaerde et al. 2023; Kim et al. 2020; Yoo et al. 2018), and this is mirrored by the perception of HCPs (Eeltink et al. 2018). Factors which have been proposed to facilitate discussion by HCP are (1) knowledge of sexual difficulties faced in this setting, (2) sufficient time for discussion, and (3) the use of a care plan to prompt discussion (Eeltink et al. 2018).

The following screening questions have been proposed (Syrjala et al. 2021):

1. Do you have any concerns regarding your sexual function, sexual activity, sexual relationship, or sex life? Yes/No
2. Are these concerns causing you distress, or would you like to discuss them? Yes/No.

If these two questions are affirmative, a more complete evaluation should be undertaken, and multimodal intervention is recommended (Syrjala et al. 2021) This approach is also endorsed by the American Society of Clinical Oncology (ASCO) and the National Comprehensive Cancer Network in the USA (NCCN) for adult patients diagnosed with cancer (Carter et al. 2018; Denlinger et al. 2017). In transplant recipients, medical problems should first be identified and treated. For example, hormone deficiencies should be addressed, and some male patients benefit from the prescription of erectile dysfunction medication. Women with vaginal dryness may benefit from lubricants or topical estrogens, and those with GVHD may benefit from topical steroids (Tirri et al. 2015). Vaginal dilators or a low-dose estradiol vaginal ring may be helpful if vaginal stenosis has occurred. Subsequently, individual and/or couple counseling should be provided as it is recognized that psychological factors can play a large part in sexual dysfunction after transplant both for the

patient and for their partner. In transplant patients, a pilot study of multimodal intervention for patients with sexual dysfunction causing distress showed promising results leading to significant improvements in the following parameters: the number of patients who were sexually active, satisfaction with and interest in sex, erectile function, orgasm, and vaginal comfort (El-Jawahri et al. 2018).

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Part VII

Prevention and Management of Relapse

Topic Leaders: Nicolaus Kröger and Peter Bader